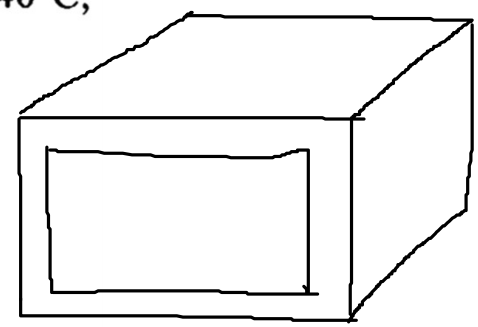
1. A furnace with a steel door, having an inner lining of an insulating material, is at a temperature of 65°C. The door, 1.5 m high and 1 m wide, loses heat to an ambient at 25°C. Calculate the rate of heat loss from the door at steady state.

SOLUTION The average free convection heat transfer coefficient $\bar{h}$ can be calculated using Eq. (5.3). The Rayleigh number will be calculated first in order to check the flow regime.

The average air-film temperature, $T_f = (65 + 25)/2 = 45°\text{C} = 318\ \text{K}$

The relevant properties of the air at this temperature are: Prandtl number, Pr = 0.695; $\nu = 1.85 \times 10^{-5}\,\text{m}^2/\text{s}$; $\beta = 1/T_f = 1/318 = 3.145 \times 10^{-3}\ \text{K}^{-1}$; $k = 0.028$ W/m °C.

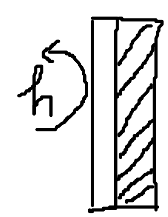
Other relevant data are: door height, $L = 1.5$ m; temperature drop, $T_s - T_o = 65 - 25 = 40°\text{C}$; $g = 9.8\ \text{m/s}^2$.



$$\text{Gr}_L = \frac{g\beta(T_s - T_o)L^3}{\nu^2} = \frac{(9.8)(3.145 \times 10^{-3})(40)(1.5)^3}{(1.85 \times 10^{-5})^2} = 1.216 \times 10^{10}$$

The Rayleigh number, $\text{Ra}_L = (\text{Gr}_L)\ (\text{Pr}) = (1.216 \times 10^{10})(0.695) = 8.45 \times 10^9$

From Eq. (5.3), $$\overline{\text{Nu}}_L = \frac{\bar{h}L}{k} = \left[0.825 + \frac{0.387(\text{Ra}_L)^{1/6}}{\left\{1 + (0.492/\text{Pr})^{9/16}\right\}^{8/27}}\right]^2$$



$$= \left[0.825 + \frac{0.387(8.45 \times 10^9)^{1/6}}{\left\{1 + (0.492/0.695)^{9/16}\right\}^{8/27}} \right]^2$$

$$= 238.4$$

The average heat transfer coefficient,

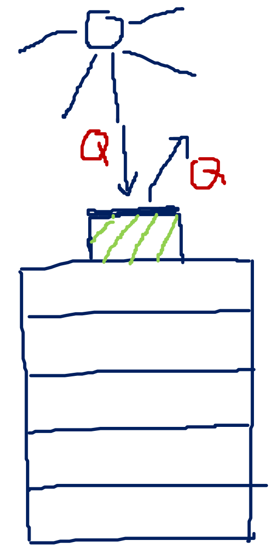
$$\bar{h} = (238.4)\left(\frac{k}{L}\right) = \frac{238.4 \times 0.028}{1.5} = 4.45 \text{ W/m}^2 \text{ °C}$$

$$\text{Door area} = (1.5)(1.0) = 1.5 \text{ m}^2$$

$$\text{Temperature driving force, } \Delta T = 65 - 25 = 40\text{°C}$$

$$\text{Rate of heat loss} = \bar{h} A \, \Delta T = (4.45)(1.5)(40) = \boxed{267 \text{ W}}$$

2. A thin metal plate 1 m × 1 m is placed on a rooftop. It receives radiant heat from the sun directly at the rate of 170 W/m^2. If heat transfer from the plate to the ambient occurs purely by free convection, calculate the steady state temperature of the plate. Assume that there is no heat loss from the bottom of the plate. The ambient temperature is 25°C.



SOLUTION At steady state, the rate of radiant heat input to the plate must be equal to the rate of heat loss to the ambient at 25°C by free convection. Thus the problem reduces to: What should be the plate temperature in order that the free convection heat flux is 170 W/m^2? We need to calculate the Grashof number in order to calculate the heat transfer coefficient, and we must know the plate temperature in order to calculate the Grashof number. But the plate temperature is what we are to find out. So the problem has to be solved by trial, starting with a guess value of the plate temperature.

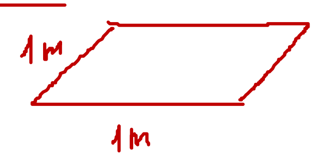
Let us assume a plate temperature, T_s = 60°C.

The mean film temperature = (60 + 25)/2 = 42.5°C = 315.5 K

Given: $T_s - T_o$ = 60 – 25 = 35°C; Properties of the air at the mean film temperature: β = 1/315.5 = 3.17×10^{-3} K^{-1}; $\nu = 1.85 \times 10^{-5}$ m^2/s; Pr = 0.7; k = 0.028 W/m °C, and g = 9.8 m/s^2.

For free convection from a horizontal plate, the characteristic length L is taken as

$$L = \frac{\text{Plate area}}{\text{Perimeter}} = \frac{1\ \text{m}^2}{4\ \text{m}} = 0.25\ \text{m}$$



$$\mathrm{Ra}_L = (\mathrm{Gr}_L)\,(\mathrm{Pr})$$

$$= \frac{g\beta(T_s - T_o)L^3}{\nu^2}\,(\mathrm{Pr}) = \frac{(9.8)(3.17 \times 10^{-3})(35)(0.25)^3}{(1.85 \times 10^{-5})^2}\,(0.70) = 3.475 \times 10^7$$

We will use Eq. (5.5) for calculating the Nusselt number though the value of Ra_L is a little out of the range.

$$\overline{\mathrm{Nu}}_L = \frac{\bar{h}L}{k} = 0.54\,(\mathrm{Ra}_L)^{1/4} = (0.54)(3.475 \times 10^7)^{1/4} = 41.46$$

or

$$\bar{h} = \frac{(41.46)(k)}{L} = \frac{(41.46)(0.028)}{0.25} = 4.64\ \mathrm{W/m^2\,°C}$$

Rate of heat transfer from the plate = $(4.64)(1.0)(T_s - 25) = 170\ \mathrm{W/m^2}$

Therefore,

$$\text{Plate temperature, } T_s = \boxed{61.6°\mathrm{C}}$$

The calculated temperature is very close to the initial guess and therefore no further trial is necessary.

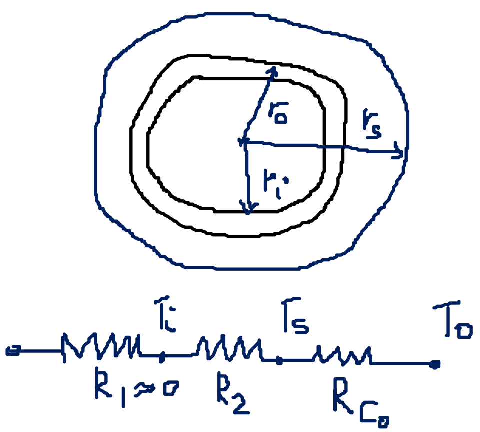
3. A horizontal steam pipe, 78 mm i.d. and 89 mm o.d., carries saturated steam at 15 kg/cm^2 gauge pressure. It is lagged only with a 2 cm thick layer of pre-formed mineral fibre of thermal conductivity 0.05 W/m °C which is exposed to ambient air at 25°C. Calculate the rate of heat loss by free convection per metre length of the pipe.

Now we calculate the convection film coefficient. The properties of the air taken at the mean film temperature, i.e. $T_f = (50 + 25)/2 = 37.5°C$ (= 310.5 K), are: $\nu = 1.76 \times 10^{-5}$ m²/s; $\beta = (1/310.5) = 3.22 \times 10^{-3}$ K^{-1}; $k = 0.027$ W/m °C; Pr = 0.705.

Saturated steam $P = 15.72$ bar, $T_i = 473.5 K$ (from steam table)

$= 1572$ kPa

Assume $T_s = 50°C$ (Trial 1)



$$Q_2 = Q_{C_o}$$

$$\frac{T_i - T_s}{R_2} = \frac{T_s - T_o}{R_{C_o}}$$

$$R_2 = \frac{\ln r_s/r_o}{2\pi L k_2}$$

$$R_{C_o} = \frac{1}{A_o h_o}$$

$$A_o = 2\pi r_s . L$$

$$h_o = ?$$

$$T_o = 25°C$$

$r_i = \frac{78}{2} = 39$ mm

$r_o = \frac{89}{2} = 44.5$ mm

$r_s = r_o + e = 64.5$ mm

$e = 2$ cm $= 20$ mm

$L = 1$ m

$k_2 = 0.05$ W/m.°C

$\Delta T = \text{constant}$

$$Q_2 = \frac{(T_i - T_s) 2\pi k_2 L}{\ln r_s/r_o} = \frac{(473.5 - 323)\, 2 \times 3.14 \times 0.05}{\ln \frac{0.0645}{0.0445}}$$

$Q_2 = 127.4\ W$

Calculate h_o (free convection)

$$Gr = \frac{\beta . g (T_s - T_o) d_s^3}{\nu^2}$$

$d_s = 2r_s = 129\,mm = 0.129\,m$

$$Gr = \frac{3.22 \times 10^{-3} \times 9.81 (50 - 25) \times 0.129^2}{(1.76 \times 10^{-5})^2}$$

$Gr = 5.47 \times 10^6$

$Ra = Gr . Pr = 3.85 \times 10^6$

<u>Geometry</u> · horizontal cylinder

$$\overline{Nu} = \frac{\bar{h} d}{k} = \left[0.60 + \frac{0.387 (Ra_d)^{1/6}}{\{1 + (0.559/Pr)^{9/16}\}^{8/27}} \right]^2$$

$$\frac{h_o d_s}{k} = 21.34$$

$\Rightarrow h_o = 4.47\ W/m^2.°C$

$Q_{co} = A_o h_o (T_s - T_o) = Q_2 = 127.4$

$$T_s - T_o = \frac{127.4}{2\pi r_s L \times h_o}$$

$\Rightarrow \boxed{T_s = 95.3°C}$

Trial 2 . Assume $T_s = 70°C$

$$T_f = \frac{70+25}{2} = 47.5°C = 320.5k$$

Air properties $\beta = \frac{1}{T_f} = 3.12 \times 10^{-3}$

$\nu = 1.8 \times 10^{-5}\ m^2/s$, $P_r = 0.703$

$k_{air} = 0.0275\ W/m.°C$

$$R_a = Gr.P_r = \frac{\beta g (T_s - T_o) d_s^3}{\nu^2} \times P_r$$

$$= 6.4 \times 10^6$$

$$\overline{Nu} = \frac{\bar{h} d}{k} = \left[0.60 + \frac{0.387(Ra_d)^{1/6}}{\{1 + (0.559/Pr)^{9/16}\}^{8/27}}\right]^2$$

$$Nu = \frac{h_o d_s}{K} = 24.97$$

$$\Rightarrow h_o = 5.32\ W/m^2.°C$$

$$Q_{c_o} = A_o h_o (T_s - T_o) = Q_2$$

$$\Rightarrow T_s = 76.4°C$$

Trial 3 . Assume $T_s = 74°C$

$$\Rightarrow \boxed{T_s = 73.9°C}$$

Ans. $T_s = 74°C$

$$\Rightarrow Q_2 = \frac{(T_i - T_s) 2\pi k_2 L}{\ln r_s/r_o} = 107\ W$$

4. A horizontal section of an uninsulated pipe, 60 mm o.d., carrying warm water runs below an exhaust fan in a factory shed. The fan creates a suction that causes a mild upward cross-flow velocity of 0.3 m/s on the pipe. If the pipe wall temperature is 60°C, calculate the rate of heat loss by the combined free and forced convection per metre length of the pipe. The ambient air temperature is 30°C.

SOLUTION Mean air-film temperature $= \dfrac{60 + 30}{2} = 45°C = 318\ K$

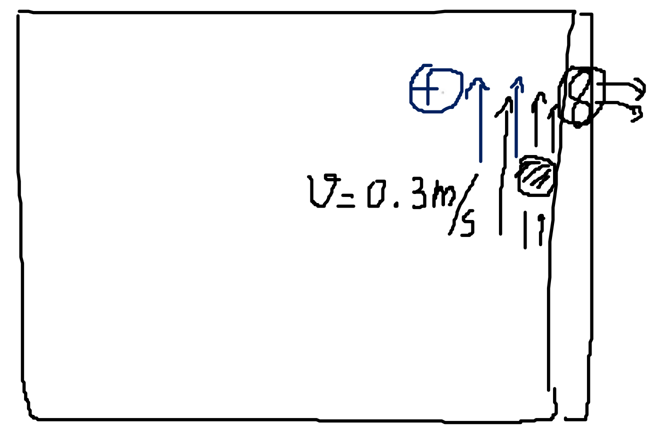
For air at 318 K, we have

$\rho = 1.105\ kg/m^3$; $c_p = 0.24$ kcal/kg °C; $\mu = 1.95 \times 10^{-5}$ kg/m s; Pr = 0.7; $\nu = \mu/\rho = 1.85 \times 10^{-5}\ m^2/s$; $k = 0.0241$ kcal/h m °C; $\beta = 1/318 = 3.145 \times 10^{-3}\ K^{-1}$; and cross flow air velocity, $V = 0.3$ m/s.

Surface temperature of the pipe, $T_s = 60°C$

Bulk temperature of air, $T_o = 30°C$

Diameter of the pipe, $d = 60$ mm $= 0.06$ m



Calculation of the free convection Nusselt number

$Re = 972$

$Gr = 5.828 \times 10^5$

$\dfrac{Gr}{Re^2} = 0.61 \simeq 1$

$$Ra_d = (Gr_d)(Pr) = \frac{g\beta(T_s - T_o)\,d^3}{\nu^2}(Pr)$$

$$= \frac{(9.8)(3.145 \times 10^{-3})(60 - 30)(0.06)^3}{(1.85 \times 10^{-5})^2}(0.7)$$

$$= 4.08 \times 10^5$$

Using Eq. (5.9), we get

$$\overline{\mathrm{Nu}}_{\mathrm{free}} = \left[0.60 + \frac{(0.387)(4.08 \times 10^5)^{1/6}}{\left\{1 + (0.559/0.7)^{9/16}\right\}^{8/27}}\right]^2 = 11.31$$

Calculation of the forced convection Nusselt number

Using the Churchill and Burnstein correlation, Eq. (4.19), we get

$$\mathrm{Re} = \frac{dV}{\nu} = \frac{(0.06)(0.3)}{1.85 \times 10^{-5}} = 972$$

$$\overline{\mathrm{Nu}}_{\mathrm{forced}} = 0.3 + \frac{(0.62)(972)^{1/2}(0.7)^{1/3}}{\left[1 + (0.4/0.7)^{2/3}\right]^{1/4}}\left[1 + \left(\frac{972}{2.82 \times 10^5}\right)\right]^{4/5} = 15.8$$

We use Eq. (5.20) to calculate the Nusselt number for mixed convection [the positive sign should be used in the expression on the RHS of Eq. (5.20), because the air flows upwards and free and forced convection processes are augmenting each other].

$$\overline{\mathrm{Nu}}^3 = \left(\overline{\mathrm{Nu}}_{\mathrm{forced}}\right)^3 + \left(\overline{\mathrm{Nu}}_{\mathrm{free}}\right)^3 = (15.8)^3 + (11.31)^3 = 3944 + 1447 = 5391$$

or

$$\overline{\mathrm{Nu}} = 17.5 = \frac{hd}{k}$$

The combined heat transfer coefficient, therefore, is

$$h = \frac{(17.5)(0.0241)}{0.06} = 7.03 \text{ kcal/h m}^2 \text{ °C}$$

The rate of heat loss per metre length of the pipe

$$= h\ A\ \Delta T = (7.03)(\pi)(0.06)(1)(60 - 30) = \boxed{39.7 \text{ kcal/h}}$$